

Day 7 Problem Set

Geometry Summer Credit | Quadrilaterals and Coordinate Tests

Name:	Date:	Period:
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Goal

Practice special quadrilateral properties and coordinate evidence using slope, distance, and midpoint.

Support Practice

1.	Find the slope between (2, 4) and (8, 7).
2.	Find the distance between (0, 0) and (5, 12).
3.	Find the midpoint of (-4, 6) and (8, 2).
4.	Solve $3x + 7 = 28$.
5.	Find x if two congruent segments are $5x - 2$ and 18.

Core Practice: Quadrilateral Properties

6.	Parallelogram ABCD has $AB = 12$ and $BC = 7$. Find CD and AD.
7.	Parallelogram ABCD has angle $A = 74$ degrees. Find angle C and angle B.
8.	A rectangle has diagonals $2x + 6$ and 34. Find x .

9. A rhombus has perimeter 48. Find each side length.

10. A square has perimeter 36. Find each side length.

11. Consecutive angles of a parallelogram are $x + 30$ and $2x$. Find x and both angles.

12. A parallelogram has opposite sides $4x - 1$ and 27. Find x .

13. A rectangle has diagonals $5x - 8$ and $3x + 10$. Find x .

Day 7 Problem Set

Continued

Core Practice: Coordinate Tests

14. Find the slopes of AB and CD for A(0, 0), B(4, 2), C(7, 6), D(3, 4). Are AB and CD parallel?

15. Find the slopes of BC and AD for A(0, 0), B(4, 2), C(7, 6), D(3, 4). Are BC and AD parallel?

16. Use #14 and #15: Is ABCD a parallelogram? Explain.

17. Find the side lengths of P(0, 0), Q(6, 0), R(6, 6), S(0, 6). Classify the quadrilateral.

18. Find the side lengths of A(0, 0), B(8, 0), C(8, 5), D(0, 5). Classify the quadrilateral.

19. Find the midpoint of diagonal AC for A(-2, 3) and C(6, 7).

20. Find the midpoint of diagonal BD for B(1, 8) and D(3, 2). Do the diagonals bisect each other?

21. A quadrilateral has both pairs of opposite sides parallel and diagonals congruent. What special quadrilateral could it be?

22. Error analysis: A student says a quadrilateral with one pair of parallel sides must be a parallelogram. Correct the statement.

23.	Spiral: Can 5, 9, and 15 form a triangle? Explain.
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24.	Challenge: What coordinate evidence would convince you a quadrilateral is a square?
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